Name: **ANSWER KEY** _____ Date: _____

Coordinate Geometry & Transformations: Color By Number

Color Code:

10 Gray (−5, 2)	**5** Red (−4, 1)	**3** Brown (4, 5)	**6** Yellow (1, −2)	**1** Light Blue (6, 2)
8 Black (4, 1)	**9** Purple (2, 3)	**2** Orange (3, 2)	**4** Green (−3, 2)	**7** Tan (−4, 10)

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Directions: Solve each problem and show your work in the box. Your answer should match one of the answers in the Color Code on the coloring sheet. Find the matching answer, write the problem number on its blank, then write the color on the line. Color every part of the picture with that number. Watch out: the answers look alike!

1.
Translate the point (2, 3) using the rule (x, y) → (x + 4, y − 1). Find the image.

(2 + 4, 3 − 1)
(6, 2)

1. Color: **Light Blue**

2.
Reflect the point (3, −2) across the x-axis. Find the image.

reflecting across the x-axis changes the sign of y
(3, 2)

2. Color: **Orange**

3.
Reflect the point (−4, 5) across the y-axis. Find the image.

reflecting across the y-axis changes the sign of x
(4, 5)

3. Color: **Brown**

4.
Rotate the point (2, 3) 90° counterclockwise about the origin. Find the image.

rule: (x, y) → (−y, x)
(−3, 2)

4. Color: **Green**

5.
Rotate the point (4, −1) 180° about the origin. Find the image.

rule: (x, y) → (−x, −y)
(−4, 1)

5. Color: **Red**

6.
Dilate the point (3, −6) by a scale factor of $\frac{1}{3}$ with the center at the origin. Find the image.

multiply both coordinates by $\frac{1}{3}$
(1, −2)

6. Color: **Yellow**

7.
Dilate the point (−2, 5) by a scale factor of 2 with the center at the origin. Find the image.

multiply both coordinates by 2
(−4, 10)

7. Color: **Tan**

8.
Reflect the point (1, 4) across the line y = x. Find the image.

reflecting across y = x swaps x and y
(4, 1)

8. Color: **Black**

9.
The translation (x, y) → (x − 3, y + 2) moves point A to A′. If A = (5, 1), find A′.

(5 − 3, 1 + 2)
(2, 3)

9. Color: **Purple**

10.
Rotate the point (−2, −5) 90° clockwise about the origin. Find the image.

rule: (x, y) → (y, −x)
(−5, 2)

10. Color: **Gray**